

SIM800A Advanced Engineering Guide

This document explains GSM communication, RF transmission, modulation, and power behavior at an engineering level.

1. GSM Architecture

GSM consists of Mobile Station (SIM800A), Base Transceiver Station (BTS), Base Station Controller (BSC), and Core Network. The SIM800A acts as a Mobile Station communicating via RF with the BTS.

2. GSM Frame Structure

Each GSM frame = 4.615 ms. It contains 8 time slots. Each time slot = 577 microseconds. A burst occupies one slot and contains training sequence, data bits, guard period, and tail bits.

3. Burst Structure

A normal GSM burst contains 148 bits: 114 data bits, 26 training bits, 8.25 guard bits, and tail bits. The guard time prevents overlap between adjacent time slots.

4. Modulation - GMSK

GMSK (Gaussian Minimum Shift Keying) is a continuous-phase FSK. Bit rate = 270.833 kbps, frequency deviation = 67.7 kHz. Gaussian filtering ($BT=0.3$) reduces spectral width but introduces controlled inter-symbol interference.

5. RF Signal Chain

Digital data → channel coding → interleaving → burst formatting → GMSK modulation → upconversion to 900 MHz → power amplification → antenna radiation.

6. RF Power Amplifier

The PA boosts signal to ~2W output. Efficiency ~40-60%. Input power ~4–8W from battery. Remaining power dissipated as heat.

7. Impedance Matching

Antenna system must be 50 ohms. Mismatch causes reflected power (VSWR). Reflection reduces transmitted power and can damage PA.

8. Free Space Path Loss

Power density decreases with square of distance: $1/(4\pi r^2)$. Received signals at tower are typically -90 dBm to -110 dBm.

9. Power Profile

Idle: 10–30 mA Search: 30–80 mA Transmit bursts: 1–2 A Voltage: ~4V Peak power: ~8W input, 2W RF output

10. Power Supply Design

Use low-ESR capacitors ($>1000\mu\text{F}$). Use short traces. Avoid long wires. Voltage drop during burst must be minimized.

11. Timing and Synchronization

GSM requires precise timing alignment with BTS. Timing advance ensures bursts arrive in correct slot.

12. Error Control

Convolutional coding and interleaving are used. Allows recovery of data even in noisy RF conditions.

13. Why TDMA Works

Multiple users share same frequency by dividing time. Each user transmits only in assigned slot, reducing interference.